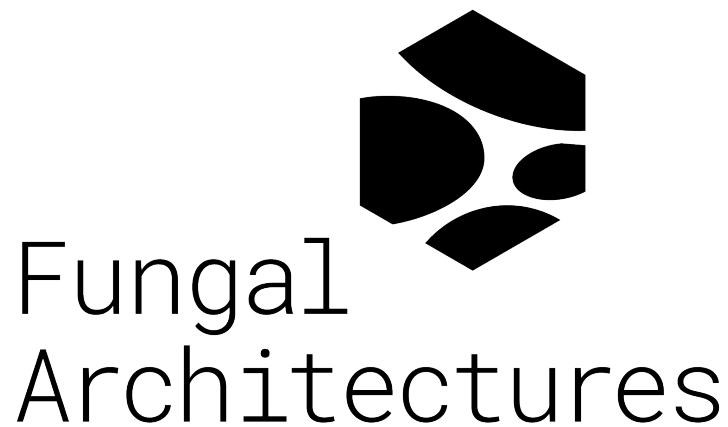


Horizon 2020



Deliverable D3.3

A map of electrical functionality of mycelium composites

Date of preparation: 30/05/2023	Revision: 1
Start date of project: 2019/12/01	Duration: 36 months
Project coordinator: UWE	Classification: public
Partners: <i>lead: UU</i>	<i>contribution: CITA, UWE, Mogu</i>
Project website:	http://fungar.eu/



H2020-FETopen-2019

DELIVERABLE SUMMARY SHEET

Grant agreement number: 858132

Project acronym: *FUNGAR*

Deliverable N°: Deliverable D3.3

Due date: M48

Delivery date: 30/05/2023

Name: **A map of electrical functionality of mycelium composites**

Description: In the deliverable we report on research activities aim to uncover electrical properties and functionalities of the mycelium composites. Advancing fungal electronics we study AC conductive properties of mycelium-bound composites and fungal fruit bodies at higher frequencies across three overlapping bands; 20 Hz to 300 kHz, 10 Hz to 4 MHz and 50 kHz to 3 GHz. Measurements indicate that mycelium-bound composites typically act as low-pass filters with a mean cut-off frequency of ~ 500 kHz; with ~ -14 dB/decade roll-off, and mean attenuation across the pass band of <1 dB. Fruiting bodies have between one or two orders of magnitude lower mean cut-off frequency (5 kHz to 50 kHz depending on species); with -20 dB/decade to -30 dB/decade roll-off, and mean attenuation across the pass band of <3 dB. The mechanism for the frequency-dependant attenuation is uncertain however the high water content ($\sim 80\%$ in mycelium-bound composites and $\sim 92\%$ in fruiting bodies depending on species) which is electrically conductive due to dissolved ionisable solids is probably a key factor. The potential for mycelium-bound composites and fruiting bodies in analog computing is explored.

Partners owning: **UU**Partners contributed: **CITA, UWE, Mogu**

Made available to: public

TABLE OF CONTENTS

1	Background	4
2	Methods and materials	4
2.1	Electrical properties of mycelium-bound composites and fungal fruiting bodies over the 20 Hz to 300 kHz frequency range.	4
2.2	Electrical properties of mycelium bound composites and fungal fruiting bodies over the 10 Hz to 4 MHz frequency range.	5
2.3	Electrical properties of mycelium bound composites and fungal fruiting bodies over the 50 kHz to 3 GHz frequency range.	6
3	Results	6
3.1	Electrical properties of mycelium-bound composites and fungal fruiting bodies over the 20 Hz to 300 kHz frequency range.	6
3.2	Electrical properties of mycelium-bound composites and fungal fruiting bodies over the 10 Hz to 4 MHz frequency range.	8
3.3	Electrical properties of mycelium-bound composites and fungal fruiting bodies over the 50 kHz to 3 GHz frequency range.	9
4	Conclusions	14
	References	15

1 BACKGROUND

Mycelium-bound composites and fungal fruit bodies have been shown to have complex electrical properties at lower frequencies including resistive spikes [2], memfractive [7, 3] and modify frequencies of external electrical inputs [15]. Electrical properties are affected by moisture content [13]. They are also known to respond to chemical and physical stimuli by changing patterns of their electrical activity [12, 1, 5, 4] and electrical conductivity [6]. In [3] we proposed a novel technological direct – fungal electronics. Fungal electronics is a family of living electronic devices made of mycelium bound composites or pure mycelium. Fungal electronic devices are capable of changing their impedance and generating spikes of electrical potential in response to external control parameters. Fungal electronics can be embedded into fungal materials and wearables or used as stand alone sensing and computing devices. Elementary fungal-electronic components must communicate with each. One of the ways to transmit information in a fault-tolerant manner is to encode it in frequencies of alternating current.

As part of our research into the information-transmissive properties of mycelium-bonded composites and fruiting bodies, we decided to study how their electrical properties change across the higher frequency spectrum [16, 17].

2 METHODS AND MATERIALS

100 g block of substrate (Rye grain seeds) well colonised with *Pleurotus ostreatus* (Ann Miller's Speciality Mushrooms, UK, [10]) was enclosed in polypropylene bags fitted with 0.5 μm air filter patch, see Fig. 2 a. The block was kept at ambient room temperature (18 °C to 22 °C) inside a growth tent (in darkness) when not being used in experiments.

Five species of fresh fruiting bodies were sourced from a local supplier (Wholesale Fruit Centre Bristol, UK), see Fig. 1 a,b,c,d,e. Two additional species were sourced from a woodland mushroom farm (Livesey Brothers Ltd, Leicestershire, UK), see Fig. 1 f,g. The fungal fruiting bodies were kept at ambient room temperature (18 °C to 22 °C) and initial measurements made within 8 h of purchase.

2.1 Electrical properties of mycelium-bound composites and fungal fruiting bodies over the 20 Hz to 300 kHz frequency range.

To make electrical connections to the mycelium-bound composites and fungal fruiting bodies, bespoke electrodes were developed. The copper crimps at the terminal end of platinum iridium needle electrodes (Technomed Ltd, UK) were soldered to the centre conductor of SMA (SubMiniature version A) right angle connectors, see Fig. 2 c. The needles were inserted ~ 15 mm depth into body of grain spawn (see Fig. 2 a) and through fungal fruiting bodies of various thicknesses (see Fig. 2 b). The distance between the centres of the needle electrodes was maintained at 20 mm by a spacer, see Fig. 2 b. 'Radial' measurements were recorded with the electrodes equal distance from the centre of the cap of the fruiting body (see Fig. 2 e) while 'axial' measurements were recorded with the electrodes perpendicular to centre of the cap. 'Radial' electrodes are therefore across the cap's gills while 'axial' electrodes are aligned with the gills (see Fig. 2 d).

Measurements were recorded within a bespoke RF-shielded test chamber, see Fig. 2. The mycelium-bound composite and fruiting bodies were electrically insulated (and physically separated by ~ 5 mm) from the inside of diecast aluminium enclosure (model Hammond 1550H, 222 mm x 146 mm x 105 mm) with a polypropylene liner. Coaxial leads and connectors (including through the RF-chamber wall) were SMA type.



Figure 1: Fungal fruiting bodies (a) *Agaricus bisporus* (Portobello), (b) *Pleurotus ostreatus* (White Oyster), (c) *Hypsizygyus tessellatus* (White Shimeji), (d) *Pleurotus djamor* (Pink Oyster), (e) *Cantharellus cibarius* (Chanterelle), (f) *Pleurotus eryngii* (King Trumpet), (g) *Lentinula edodes* (Shiitake).

Electrical impedance and other circuit parameters were measured using a digital Inductance Capacitance Resistance (LCR) meter (model 891, BK Precision Ltd, UK [14]). The LCR meter was configured to scan across the 20 Hz to 300 kHz frequency range applying 1 V_{rms} sinusoidal voltage waveform through the mycelium bound composite and fruiting bodies. 301 measurements being automatically recorded per sweep with $\sim 0.05\%$ accuracy.

2.2 Electrical properties of mycelium bound composites and fungal fruiting bodies over the 10 Hz to 4 MHz frequency range.

The signal propagation was measured using an impedance – amplitude – phase frequency response network analyzer (C60, Cypher Instruments, London, UK [8]). The C60 network analyzer passes 2 V_{pp} sinusoidal voltage waveform through the mycelium bound composite and fruiting bodies at a plurality of frequencies (10 Hz to 4 MHz). The network analyzer was connected to the native CypherGraph (V1.28.0) software package on a Windows computer to control functionality and store measurements. The software evaluates the waveform after it passes through the sample and displays it as a Bode plot, the frequency response was analyzed and measurements stored.

Experimental ‘controls’ were recorded using the same setup, however, the mycelium-bound composite was replaced with; uncolonised substrate (Rye grain seeds in 50 mL glass beaker) and electrically insulating substrate (open cell polyurethane foam with water of different conductivity in 50 mL glass beaker). The ‘controls’ were subject to the same frequency spectrum of sinusoidal waveforms to explore if part of the signal was being propagated through the fungal hypha rather than the substrate or the instrumentation.

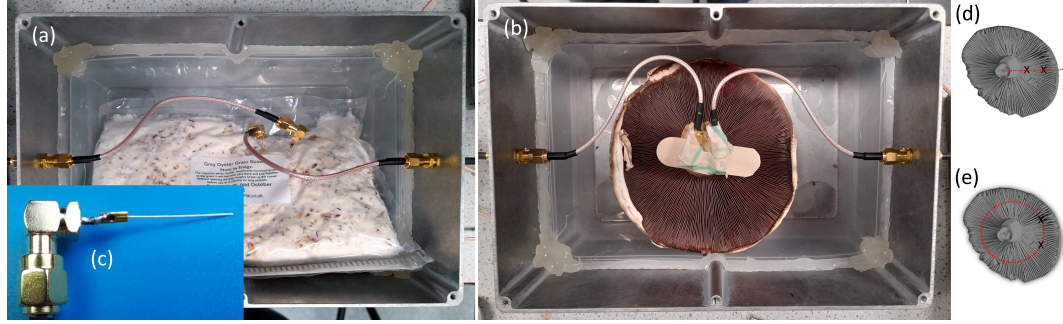


Figure 2: Electrodes inserted into (a) mycelium bound composite (Grey Oyster spawn) in bag (b) fruiting body (*Agaricus bisporus*) inside RF shielded enclosure via coaxial connections (c) platinum iridium needle electrode with SMA connector (d) axial alignment of electrodes in cap of fruiting body (e) radial alignment of electrodes in cap.

2.3 Electrical properties of mycelium bound composites and fungal fruiting bodies over the 50 kHz to 3 GHz frequency range.

S-parameters S11 and S21 were measured using the previously described setup of electrodes and RF-shielded test chamber. This allowed reflection and transmission measurements. The signal propagation was measured using a Vector Network Analyser (NanoVNA-F V2, Amazon Plc, UK [11]). The instrument's maximum output power depends on the frequency and is between -14 dBm to -19 dBm. The dynamic range for reflection measurements (S11) is 70 dB or better up to 1.5 GHz and 60 dB or better up to 1.5 GHz to 3 GHz.

The Vector Network Analyser (VNA) was regularly calibrated to maintain accuracy involving short-circuit, $50\ \Omega$ load, and open termination, followed by a direct connection between ports (inside the RF-chamber). VNA was connected to the native NanoVNA-QT VNA Saver (V 0.5.3) software package on a Windows computer to control functionality and store measurements.

3 RESULTS

3.1 Electrical properties of mycelium-bound composites and fungal fruiting bodies over the 20 Hz to 300 kHz frequency range.

The LCR meter used in this experiment fits a parallel model ($R_p\ C_p$) for low capacitance measurement and series model ($R_s\ C_s$) for high capacitance measurement. Since the capacitance of the mycelium is high, it has fitted a series RC network as shown Figure 3 with series resistance (R) and a series capacitance (C_s). The reason being if the capacitance is small, the parallel resistance becomes large and more relevant than the series resistance. If the capacitance is large, the series resistance becomes dominant and the parallel resistance becomes insignificant.

[thick,american voltages] (0,0) to [short, *-] (0.5,0) to [european resistor, l= R] ++(2,0) to [capacitor,l= C_s ,*] ++(2,0);

Figure 3: Equivalent circuit for the mycelium substrate

The electrical impedance $Z(f) = R + \frac{1}{j\omega C_s}$ of the fruiting body (cap and stalk) decreased with increasing frequency over 20 Hz to 300 kHz range. The impedance of mycelium-bound composite

also decreased but at a considerably slower rate, see Figure 4. The stalk of the fruiting body has a higher electrical impedance than the cap. The axial impedance of cap is lower than the radial impedance at lower frequencies becoming similar ~ 300 kHz. This trend shows that mycelium-bound composites demonstrate lower impedance at higher frequencies.

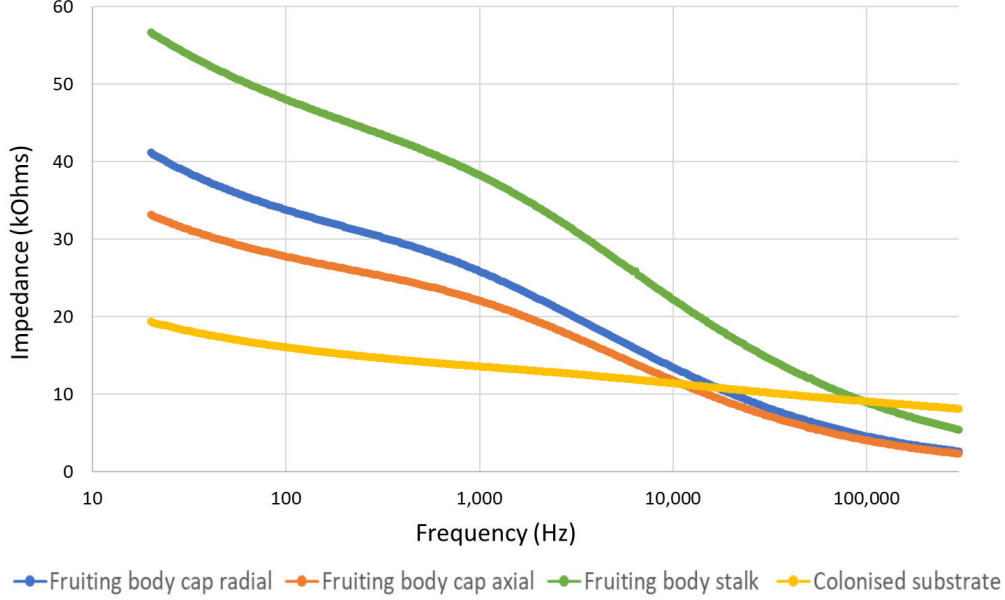


Figure 4: Impedance against frequency of mycelium-bound composites and fungal fruiting bodies (cap and stalk) over 20 Hz to 300 kHz range.

Table 1: Electrical properties of fruiting body and colonised substrate at 300 kHz

	Fruiting body			Colonised substrate
	cap		stalk	body
	radial	axial	radial	
R (k Ω)	1.991	1.782	4.438	8.312
Cs (pF)	348	394	156	551
Z (k Ω)	2.490	2.230	5.580	8.390

A summary of other electrical properties of mycelium-bound composites and fungal fruiting bodies is shown in Table 1. The fruiting body's stalk has less than half the capacitance, more than twice the inductance, and lower steady-state DC resistance of the cap (both radial and axial). The steady-state DC resistance of the colonised substrate is higher than the fruiting body.

Using the lump model of the fruiting body and the colonised substrate, we have carried out SPICE S-parameter simulation to understand the frequency dependant characteristics of the mycelium substrate for signal transfer characteristics. Figure 5 depict the S-parameter simulation of the lump model for the return loss (S11) and the gain (S21) for different cases. In very low frequencies up to 100 kHz, a significant amount of the signal is reflected back to the source and the amount of signal transferred through the substrate is very low. However, starting from

around 1 MHz onward the return loss becomes negative around -0.1 to -0.5 which is still higher in comparison to a normal semi-conductive substrates such as silicon.

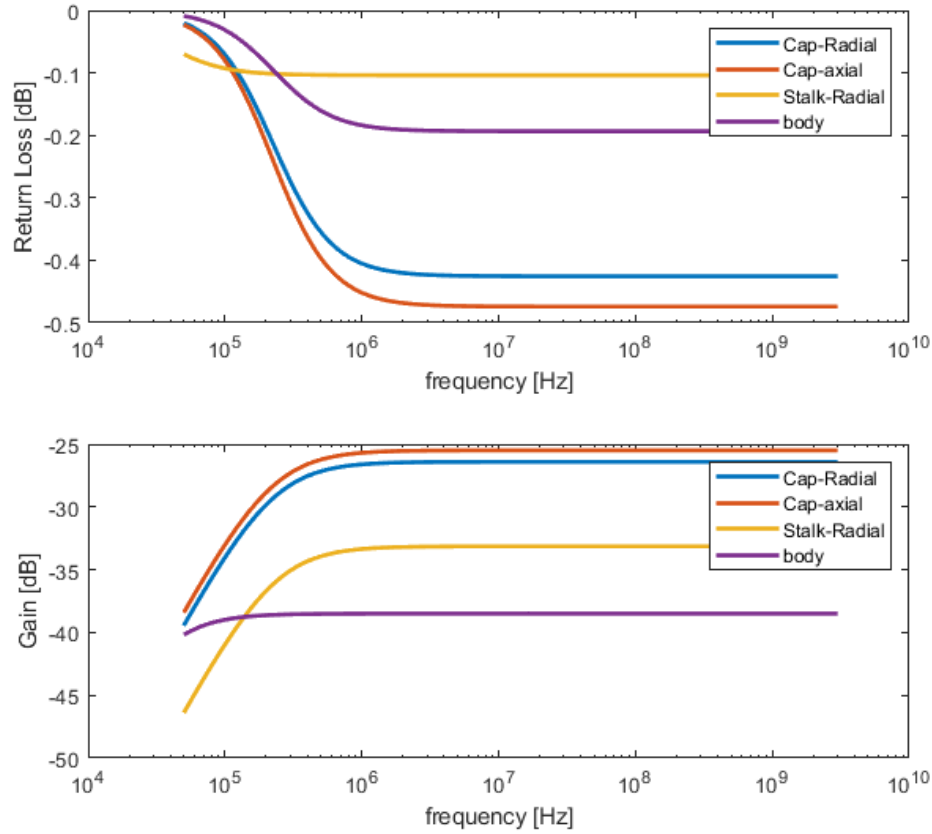


Figure 5: SPICE S-parameter simulation using the lump equivalent circuit

3.2 Electrical properties of mycelium-bound composites and fungal fruiting bodies over the 10 Hz to 4 MHz frequency range.

The signal propagation passing through mycelium-bound composite was frequency dependent; the lower frequency waveforms passed through with little attenuation while the higher frequencies became increasingly attenuated. The magnitude frequency profile matched that of a low pass filter; the phase response also appeared to correlate with a typical low pass filter. The Bode plot of mycelium-bound composite, see Figure 6, shows the level of attenuation increased noticeably above 100 kHz (e.g. -8 dB at 1 MHz). The phase decreases noticeably above 10 kHz.

The Bode plots of five species of fungal fruiting bodies are shown in Fig. 7. For measurement consistency, the electrodes (with 20 mm spacing) were normally inserted into the caps of the fruiting bodies. However, the smaller physical size of *Hypsizygus tessellatus* meant that in one recording both electrodes were inserted into the stalk, and for a second recording one electrode was in the stalk and one in the cap. The mean cut-off frequency was between 5 kHz to 50 kHz

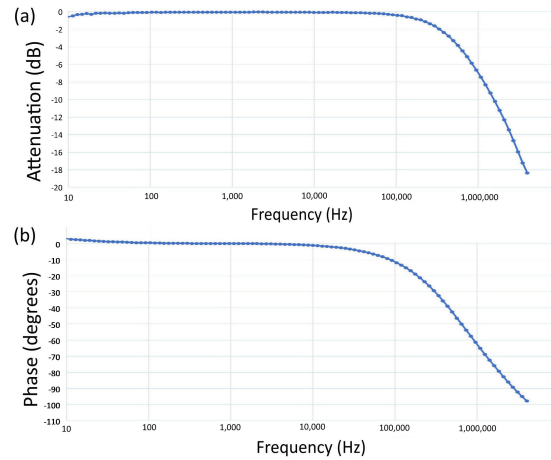


Figure 6: Bode plot of mycelium-bound composite (a) attenuation against frequency (b) phase against frequency (10 Hz to 4 MHz).

(depending on species); -20 dB/decade to -30 dB/decade roll-off, with mean attenuation across the pass band of ~ -3 dB.

To support the analysis of material properties (attenuation and phase against frequency) six configurations were measured; well colonised substrate (*Pleurotus ostreatus* on Rye seeds, blocks of 100 g and 750 g), uncolonised substrate (Rye seeds ~ 66 % moisture content and Rye seeds with 2 mL of mains water 0.0025 M Ω cm added to increase moisture content to ~ 76 %, in 50 mL glass beakers) and open cell polyurethane foam sponge (with 2 mL de-ionised water 10 M Ω cm and 2 mL mains water 0.0025 M Ω cm, in 50 mL glass beakers), see Figure 8.

It was observed during measurements that the cap and stalk of fruiting bodies can have significantly different electrical properties. By way of example, Fig.9 shows an exemplar of sequential recording from the same *Agaricus bisporus* fruiting body. It was not possible to measure some species as their stalks were physically too small to accommodate electrodes with 20 mm separation.

3.3 Electrical properties of mycelium-bound composites and fungal fruiting bodies over the 50 kHz to 3 GHz frequency range.

Signal propagation in the mycelium-bound composite is frequency-dependent. Figure 10 shows S11 the return loss against frequency of exemplar recording. Resonance peaks are observed around 960 MHz, 1.05 GHz, 1.61 GHz, and 2.04 GHz.

Gain against frequency (S21) of exemplar recording of mycelium-bound composite is shown in Figure 11. Resonance peaks in gain are observed around 64 MHz, 898 MHz, 1.17 GHz, 1.89 GHz, 1.99 GHz, and 2.70 GHz.

Return loss against frequency (S11) in cap and stalk of *Agaricus bisporus* (Portobello) fruiting bodies, is shown in Fig. 12. Signal propagation was observed to be frequency-dependent with resonance peaks around 180 MHz, 780 MHz, 1.4 GHz, 2.1 GHz and 2.7 GHz.

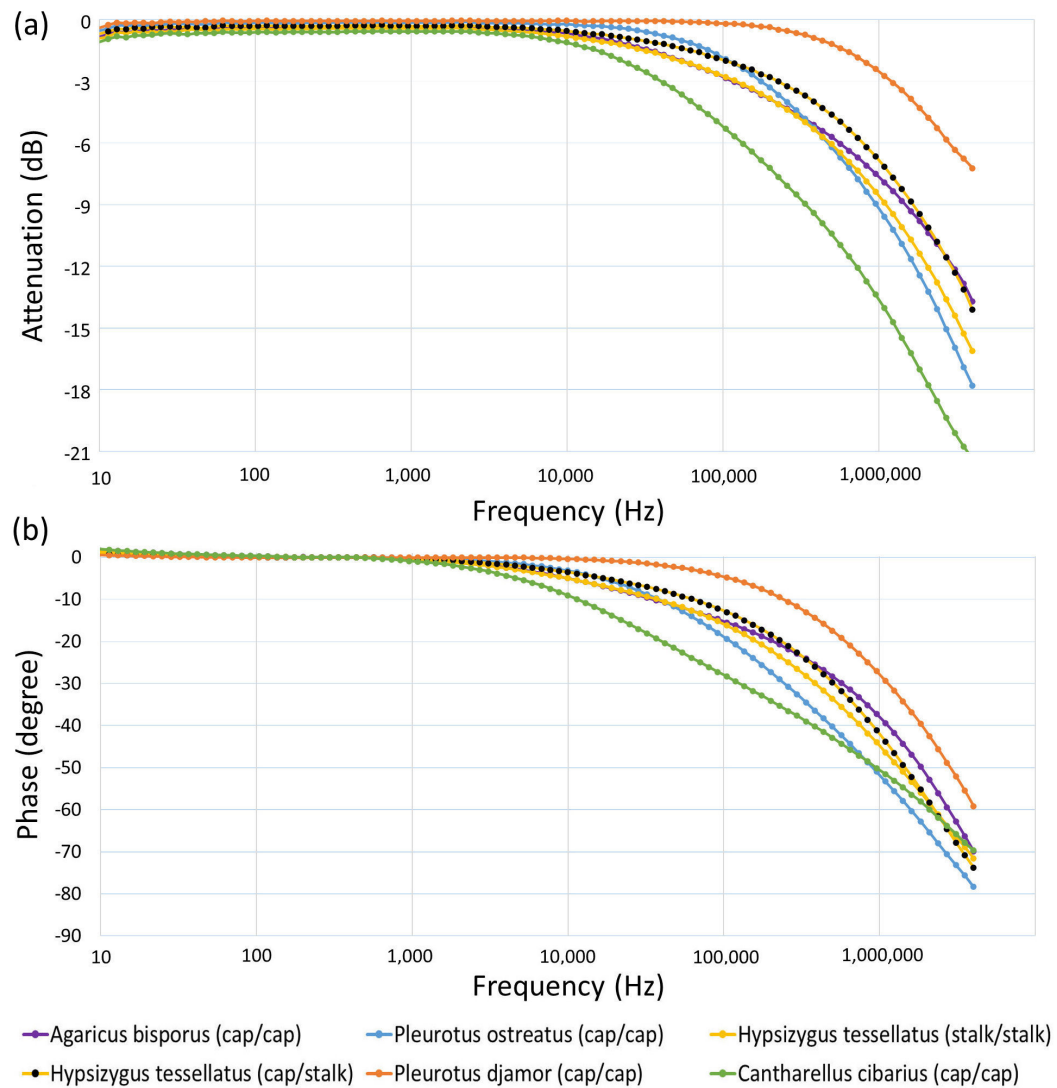


Figure 7: Bode plots of five species of fungal fruiting bodies (a) attenuation against frequency (b) phase against frequency (10 Hz to 4 MHz).

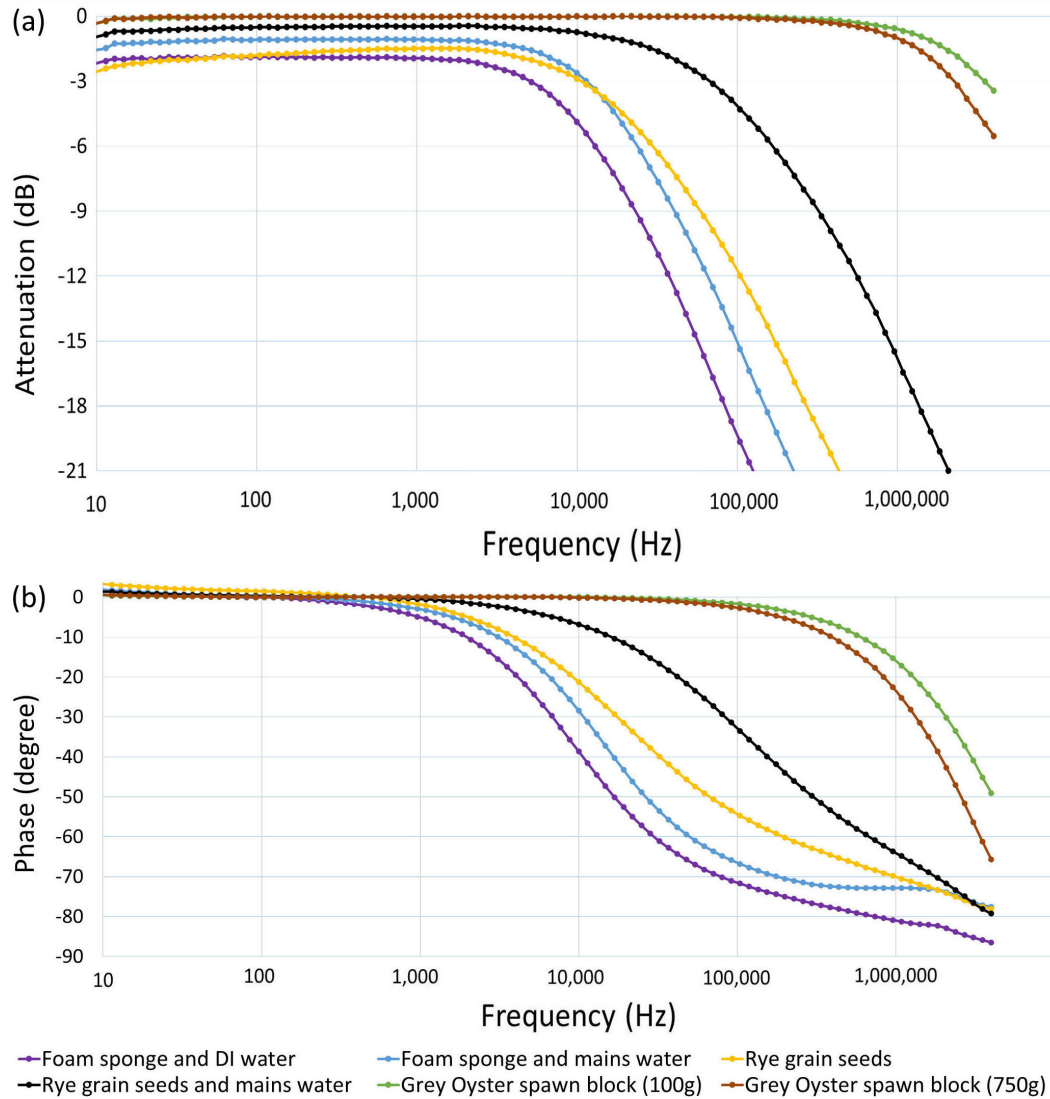


Figure 8: Bode plots of six configurations; foam sponge and de-ionised water, foam sponge and mains water, Rye grain seeds, Rye grain seeds and mains water, Grey Oyster spawn block (100 g), Grey Oyster spawn block (750 g), (a) attenuation against frequency (b) phase against frequency (10 Hz to 4 MHz).

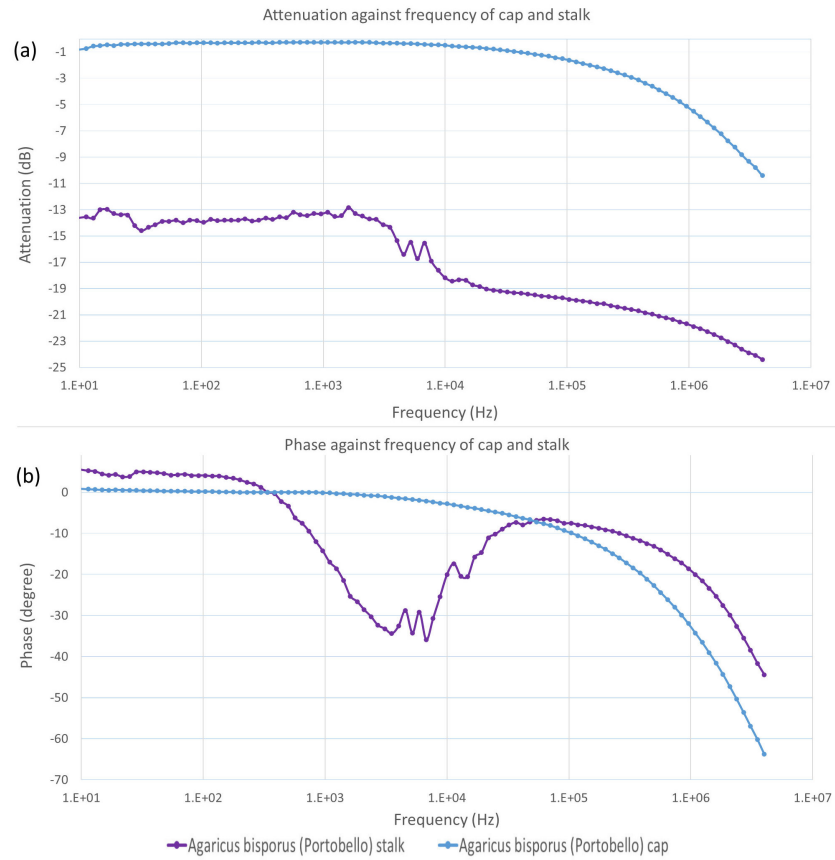


Figure 9: Bode plots of cap and stalk of *Agaricus bisporus* fruiting body (a) attenuation against frequency (b) phase against frequency (10 Hz to 4 MHz).

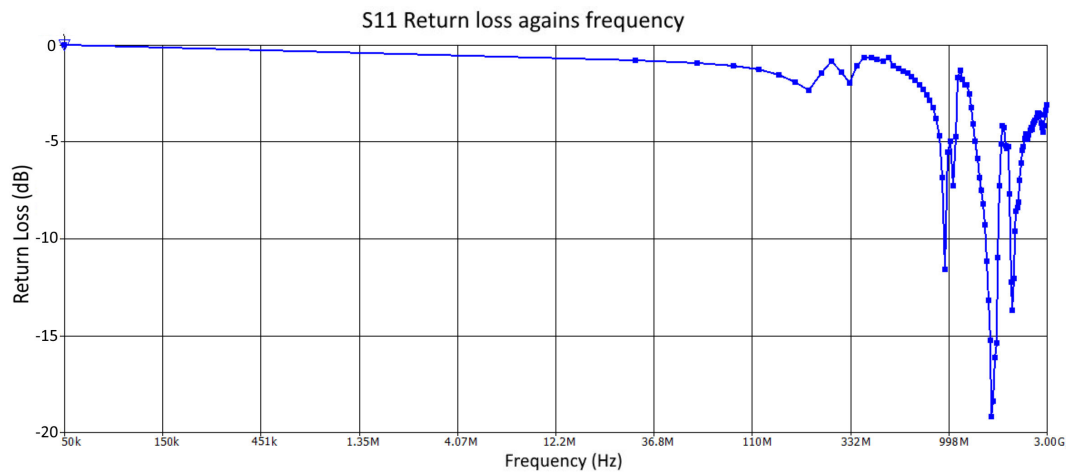


Figure 10: S11 - Return loss against the frequency (50 kHz to 3 GHz) of mycelium bound composite (Rye seeds well colonised with *Pleurotus ostreatus*).

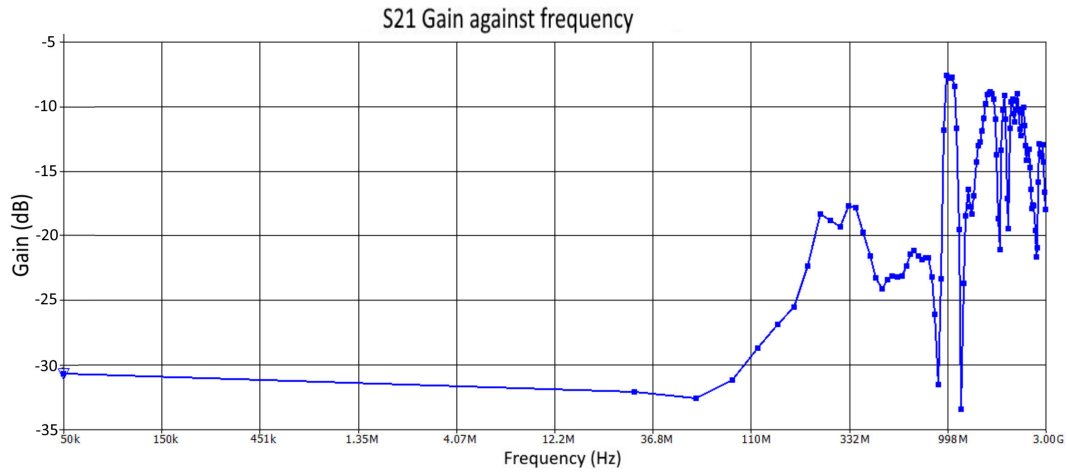


Figure 11: S21 - Gain against frequency (50 kHz to 3 GHz) of mycelium bound composite (Rye seeds well colonised with *Pleurotus ostreatus*).

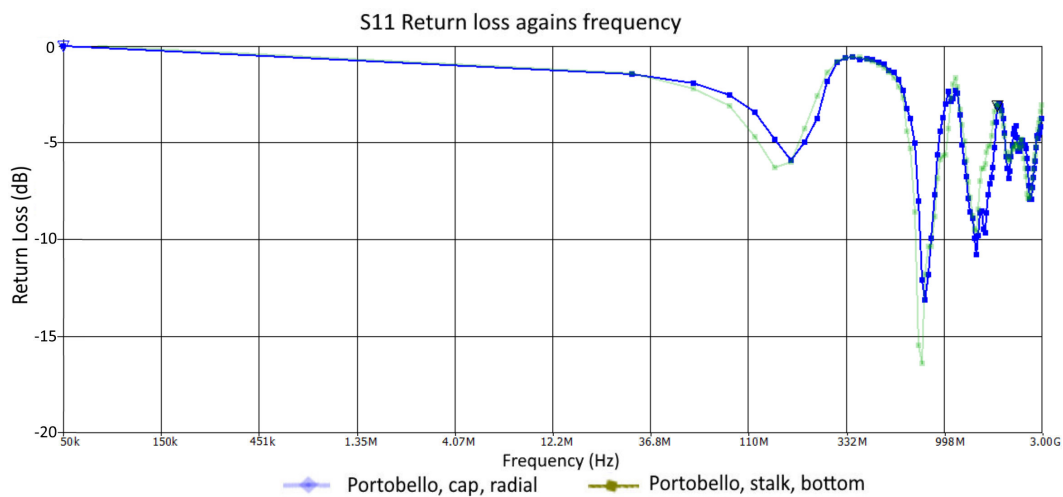


Figure 12: Return loss against frequency (50 kHz to 3 GHz) of cap and stalk of *Agaricus bisporus* fruiting body.

Signal gain against frequency (S21) in cap and stalk of *Agaricus bisporus* (Portobello) fruiting bodies is shown in Fig. 13, with peaks in attenuation around 780 MHz, 1.18 GHz, 1.89 GHz, 2.07 GHz, 2.26 GHz, and 2.76 GHz.

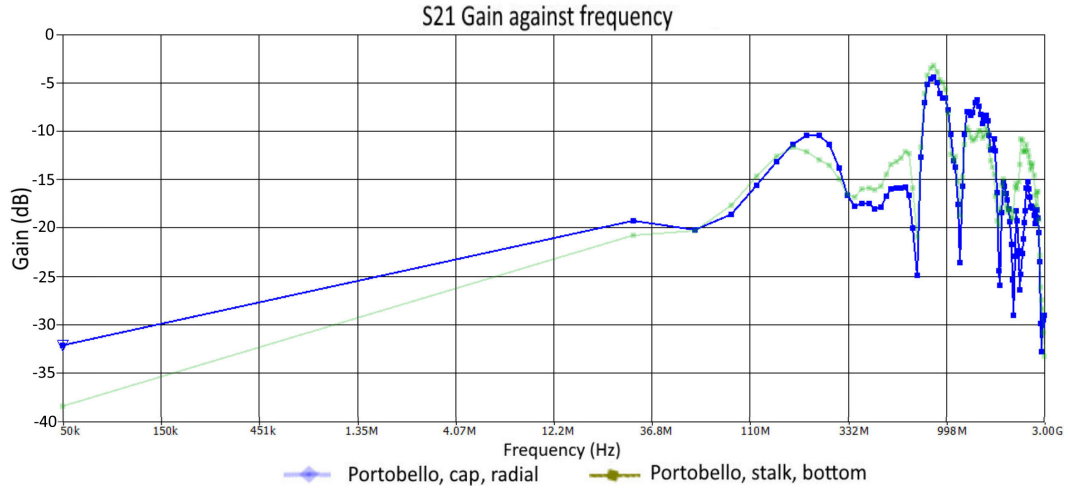


Figure 13: Gain against frequency (50 kHz to 3 GHz) of cap and stalk of *Agaricus bisporus* fruiting body.

The moisture content of mycelium-bound substrates and fruiting bodies was determined by the following procedure: (1) measure the ‘wet’ weight of the sample (2) dehydrate the sample in an oven at 80 °C for ~ 48 h (3) measure the ‘dry’ weight of the sample (4) calculate the difference between ‘wet’ and ‘dry’ weights then dividing by the ‘dry’ weight.

Overall, electrical characteristics were observed to vary with electrode separation. In other recordings (not reported in this paper) a distance of ~20 mm between centres of electrodes was found to be effective for measuring electrical properties. This suggests that there is an optimum spacing for the electrodes in any environment. Optimising the relative physical positions of electrodes in colonised substrate and fungal fruiting bodies is important to maximising the sensitivity of monitoring and interconnections to other systems.

Obtaining fresh blocks of spawn (e.g. 100 g bags) from commercial suppliers at desired times was challenging (e.g. limited stock availability). Further, most commercial suppliers were unwilling to provide details of substrate composition (beyond “Rye seeds” as considered a ‘trade secret’). Therefore, variation in substrate might exist between both batches from the same supplier and different suppliers. The level of colonisation of blocks varied greatly between suppliers and times of recordings as the fungi consumed the substrate as a source of nutrients. Additionally, the heterogeneous mixture of substrate and fungi added an additional variable.

4 CONCLUSIONS

Mycelium-bound composites was electrically characterised for signal transfer analysis with an aim to use them in electronic applications such as bio-hybrid computing systems.

Mycelium-bound composites act as low-pass filters with a mean cut-off frequency of ~500 kHz; with ~-14 dB/decade roll-off, and mean attenuation across the pass band of <1 dB. Fruiting bodies have between one or two orders of magnitude lower mean cut-off frequency (5 kHz to 50 kHz depending on species); with -20 dB/decade to -30 dB/decade roll-off, and mean attenuation

across the pass band of <3 dB. The mechanism underlying frequency-dependent attenuation is uncertain. The fine networks of hyphae may be better able to carry high-frequency signals than ‘bulk’ volumes of water (such as a wet foam sponge).

Mycelium-bound composites and fungal fruiting bodies may be useful building elements in analog computing. For example, frequency filters in Oscillator Computing and sensory data handling in Edge Computing. Living mycelium might be used for sensing, sensorial fusion, and pre-processing. Further, integrating mycelium networks and silicon components (potential on dried mycelium) might facilitate novel capabilities for the next generation of analog computing. It is evident from this study that mycelium-bound composites are very lossy low to high frequency range and demonstrates dielectric properties. Even though they can be considered as electrical insulators, with the moisture content the electrical breakdown voltage may be reduced and therefore they are not suitable for high-voltage applications. Transferring electrical signals through mycelium is quite challenging but mixing it with conductive particles would enable sustainable electronic designs in the future [9].

REFERENCES

- [1] Andrew Adamatzky. “On spiking behaviour of oyster fungi *Pleurotus djamor*”. In: *Scientific reports* 8.1 (2018), pp. 1–7.
- [2] Andrew Adamatzky, Alessandro Chiolerio and Georgios Sirakoulis. “On resistive spiking of fungi”. In: *arXiv preprint arXiv:2009.00292* (2020).
- [3] Andrew Adamatzky, Antoni Gandia and Alessandro Chiolerio. “Fungal sensing skin”. In: *Fungal biology and biotechnology* 8.1 (2021), pp. 1–6.
- [4] Andrew Adamatzky et al. “Logics in fungal mycelium networks”. In: *arXiv preprint arXiv:2112.07236* (2021).
- [5] Andrew Adamatzky et al. “Towards Cytoskeleton Computers. A proposal”. In: *From parallel to emergent computing*. Ed. by Andrew Adamatzky, Selim Akl and Georgios Sirakoulis. CRC Group/Taylor & Francis, 2019.
- [6] Alexander E Beasley, Anna L Powell and Andrew Adamatzky. “Fungal photosensors”. In: *arXiv preprint arXiv:2003.07825* (2020).
- [7] Alexander E Beasley et al. “Mem-fractive properties of mushrooms”. In: *arXiv preprint arXiv:2002.06413* (2020).
- [8] Cypher-Instruments. *C60 network analyzer*. <http://www.cypherinstruments.co.uk/>. [Online; accessed 18-Sept-2022]. 2022.
- [9] Doris Danninger et al. “MycelioTronics: Fungal mycelium skin for sustainable electronics”. In: *Science Advances* 8.45 (2022), eadd7118. DOI: [10.1126/sciadv.add7118](https://doi.org/10.1126/sciadv.add7118). eprint: <https://www.science.org/doi/pdf/10.1126/sciadv.add7118>. URL: <https://www.science.org/doi/abs/10.1126/sciadv.add7118>.
- [10] Ann Miller. *Oyster grain spawn*. <https://www.annforfungi.co.uk/shop/oyster-grain-spawn/>. [Online; accessed 18-Sept-2022]. 2022.
- [11] NanoRFE. *Vector Network Analyse, model NanoVNA-F V2*. <https://nanorfe.com/nanovna-v2.html>. [Online; accessed 18-Sept-2022]. 2022.
- [12] S Olsson and BS Hansson. “Action potential-like activity found in fungal mycelia is sensitive to stimulation”. In: *Naturwissenschaften* 82.1 (1995), pp. 30–31.

- [13] Neil Phillips, Antoni Gandia and Andrew Adamatzky. “Electrical response of fungi to changing moisture content”. In: *TBC Journal* (2022).
- [14] B. K. Precision. *Model 891*. https://www.bkprecision.pl/files/891_datasheet.pdf. [Online; accessed 18-Sept-2022]. 2022.
- [15] Dawid Przyczyna et al. “Electrical frequency discrimination by fungi *Pleurotus ostreatus*”. In: *arXiv preprint arXiv:2210.01775* (2022).
- [16] Botao Shao et al. “High frequency characterization and modelling of inkjet printed interconnects on flexible substrate for low-cost RFID applications”. In: *Electronics System-Integration Technology Conference*. 2008, pp. 695–700. DOI: [10.1109/ESTC.2008.4684435](https://doi.org/10.1109/ESTC.2008.4684435).
- [17] Botao Shao et al. “High Frequency Characterization of Inkjet Printed Coplanar Waveguides”. In: *IEEE Workshop on Signal Propagation on Interconnects*. 2008, pp. 1–4. DOI: [10.1109/SPI.2008.4558383](https://doi.org/10.1109/SPI.2008.4558383).